SQL Fundamentals : SELECT & FILTERING

Table : Employees

ID	First Name	Last Name	department	Salary	hire-date	city
1	Alice	Green	IT	70 000	2020/01/10	Joburg
2	Brian	Lee	HR	45 000	2019/03/22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018/07/18	Durban
4	David	Mokoena	Marketing	50 000	2021/11/05	Pretoria
5	Eva	Naidoo	IT	72 000	2017/09/30	Joburg
6						

Question 1 : Retrieve all columns from employees table

```
SELECT *
FROM Employees
```

ID	First Name	Last Name	department	Salary	hire-date	city
1	Alice	Green	IT	70 000	2020/01/10	Joburg
2	Brian	Lee	HR	45 000	2019/03/22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018/07/18	Durban
4	David	Mokoena	Marketing	50 000	2021/11/05	Pretoria
5	Eva	Naidoo	IT	72 000	2017/09/30	Joburg

Question 2: Find all unique departments

```
SELECT DISTINCT departments
FROM Employees;
```

Departments
IT
HR
Finance
Marketing

Question 3: Retrieve first and last names, ordered by salary descending.

```
SELECT first_name, last_name, salary
FROM Employees
ORDER BY salary DESC;
```

first_name	last_name	salary
Eva	Naidoo	72 000
Alice	Green	70 000
Cathy	Zulu	65 000
David	Mokoena	50 000
Brian	Lee	45 000

Question 4 : Retrieve top 3 highest paid employees

```
SELECT id, first_name, last_name, salary
FROM Employees
ORDER BY salary DESC
LIMIT 3;
```

ID	first_name	last_name	Salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

Question 5 : Find employees in IT department

```
SELECT id, first_name, last_name, department
FROM Employees
WHERE department = 'IT'
```

ID	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

Question 6 : Find employees in Finance with salary >60 000

```
SELECT id, first_name, last_name, department,
       Salary
FROM   Employees
WHERE  department = 'IT' and Salary >60 000;
```

ID	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65 000

Question 7 : Find employees in HR or marketing

```
SELECT id, first_name, last_name, department
FROM   Employees
WHERE  department = 'HR' or department = 'Marketing'
```

ID	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

Question 8 : Find employees NOT in IT

```
SELECT id, first-name, last-name, department
 FROM Employees
WHERE department =! 'IT' ;
```

ID	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

Question 9 : Find employees in IT, HR or Finance using IN

```
SELECT id, first-name, last-name, department
FROM Employees
WHERE department IN ('IT', 'HR', 'Finance')
```

ID	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

Question 10 : Find employees in IT with salary >65 000 and city in Joburg

```
SELECT *
FROM Employees
WHERE department = 'IT' AND salary >65 000
AND city = 'Joburg' ;
```

ID	first-name	last-name	department	salary	hire-date	city
1	Alice	Green	IT	70 000	2020/01/10	Joburg
5	Eva	Naidoo	IT	72 000	2017/09/30	Joburg